Dual-Path Learning Toward Open-Vocabulary Object Detection in Remote Sensing Images

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***Abstract*—While the landscape of closed-set oriented object detection has been revolutionized in recent years, extending this success to the instances of unseen categories, the core challenge of open-vocabulary object detection (OVD), remains a formidable hurdle. Prevailing methods often resort to a dedicated student detector to channel the cross-modal knowledge from pretrained vision-language models (VLMs), thereby enabling open-vocabulary generalization. However, these approaches, which primarily manifest as either knowledge distillation or pseudo-labeling frameworks, suffer from a fundamental tradeoff: biased region proposal network (RPN) yields poor recall for target objects, while shared detection head induces catastrophic forgetting of base classes. To overcome these limitations, we propose DPNet, a dedicated end-to-end dual-path network for open-vocabulary oriented object detection in remote sensing scenarios. First, we introduce a parameter-efficient multiscale adapter (MSA) that activates hierarchical features from a frozen vision-language encoder, enabling a category-agnostic network to generate high-quality proposals for target objects with varying scales. Second, a decoupled detection pipeline is formulated for base and target classes. This divide-and-conquer fashion alleviates the inherent optimization conflict, preserving base-class recognition fidelity while effectively learning target concepts. By circumventing the need for a student detector, DPNet offers a more streamlined and efficient architecture. Extensive experiments on two standard remote sensing benchmarks—DIOR and DOTA—validate that our solution achieves competitive performance, significantly outperforming pioneering approaches in both base-class precision and novel-class discovery under zero-shot settings. The source code will be available at https://github.com/yanqingyao1994/DPNet**

***Index Terms*—Category-agnostic oriented proposal generation (COPG), dual-path detection pipeline, multiscale adapter (MSA), open-set object detection, oriented detector, remote sensing images.**

I. Introduction

DEEP neural networks have revolutionized object detection in aerial imagery, achieving remarkable success in closed-set scenarios [1], [2], [3], [4], [5], [6], [7]. These advancements underpin critical applications ranging from ecological monitoring and urban planning to military reconnaissance [8], [9], [10], [11], [12], [13]. However, this remarkable success has been largely confined to the static world, where all object categories are predefined during training. In practice, the Earth's surface hosts an immense diversity of objects that far exceeds the narrow scope of current remote sensing datasets [14], [15], [16], [17], [18], most of which cover fewer than 20 categories. This closed-set paradigm struggles to address real-world demands, as annotating all potential categories is prohibitively expensive and logistically infeasible.

To bridge this gap, the open-vocabulary object detection (OVD) has emerged as a pragmatic paradigm: it endows detectors trained on *base* categories with the capacity to generalize to *target* categories through minimal human guidance (e.g., textual descriptions). The success of OVD in natural images largely stems from vision-language models (VLMs), where large-scale pretrained models, such as CLIP [19] and ALIGN [20], establish robust cross-modal alignment between images and text, enabling zero-shot recognition of unseen categories. Aiming at effectively channeling this cross-modal knowledge into a detection framework, two dominant strategies have emerged, differing fundamentally in how the teacher model's knowledge is integrated. Knowledge distillation-based approaches [21], [22], as illustrated in Fig. 1(a), leverage VLM to extract visual and textual features of region proposals, thereby guiding the student model's imitative learning process. Conversely, in pseudo-labeling methods [23], [24], [25], the pretrained models act as an annotator to generate pseudo-labels for class-agnostic region proposals, which are then used to train the student model and equip it with the capability to detect target categories, as exhibited in Fig. 1(b).

While the aforementioned paradigms achieve considerable efficacy, these preceding efforts pervasively struggle with inherent hurdles. On the one hand, biased by its closed-set training, the binary classifier in region proposal network (RPN) systematically dismisses target objects as background, severely crippling the recall of target categories [26]. On the other hand, the parameter-sharing design between base- and target-class detectors introduces a critical optimization conflict [27]. Specifically, competing gradient updates from target categories pollute the parameters fine-tuned for base-class recognition, culminating in a form of catastrophic forgetting in which the model's well-established knowledge of base categories is progressively eroded [28], [29]. Furthermore, the encoders in pretrained VLMs are conventionally frozen during training. While effective at preserving cross-modal alignment, such static setup hinders the model's adaptability to downstream

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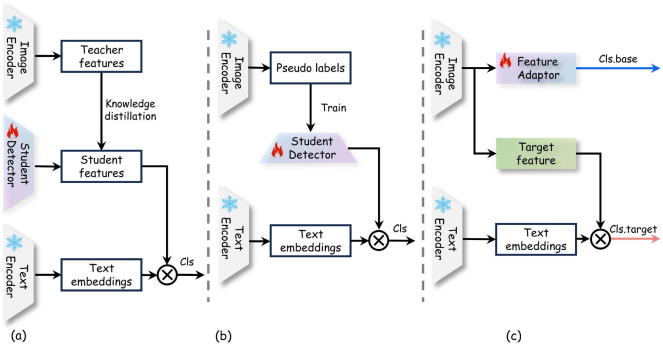


Fig. 1. Intuitive comparison between (a) knowledge distillation-based paradigm, (b) pseudo-label generation-based approaches, and (c) proposed method. Both (a) and (b) require an auxiliary student model to explicitly or implicitly acquire knowledge from the pretrained model. In contrast, our method learns to detect open-set oriented objects through a lightweight trainable base-class feature adapter coupled with a dual-path detection branch, resulting in a more streamlined and efficient architecture.

tasks. Consequently, the gains on target classes within these methods are typically achieved at the expense of performance degradation on base classes.

Worse still, these core obstacles are particularly acute within the unique context of remote sensing imagery [30], [31], [32], [33], [34]. To be specific, the intrinsic challenge toward generating high-quality proposals for objects with arbitrary orientations and dense packing configurations exacerbates the semantic bias of conventional RPN. This, in turn, impedes the model's ability to effectively encompass potential target-class objects. Moreover, the top–down perspective inherently diminishes the rich visual cues of objects, compelling recognition to rely exclusively on subtle differences in morphology and texture. This necessitates a highly customized training regime for differentiating target classes based on these subtle cues. Such specialization, however, inadvertently undermines the generalized feature representations essential for base classes and further exacerbates the risk of knowledge collapse.

Bringing the aforementioned barriers together, we propose a dual-path framework tailored to open-vocabulary oriented object detection in remote sensing images, dubbed DPNet. As depicted in Fig. 1(c), to unlock the potential of large pretrained models for downstream detection while simultaneously addressing the significant scale variation inherent in aerial scenarios, we first establish a lightweight multiscale adapter (MSA) to formulate hierarchical representations, upon which a category-agnostic oriented proposal generation (COPG) network is devised to yield high-quality candidates for target instances. Moreover, we explicitly decouple the detection of base and target classes with a dual-path architecture. This divide-and-conquer design substantially counteracts the performance degradation on base categories while delivering a robust detection on target instances.

In essence, our approach not only circumvents the need for complex knowledge distillation but also eliminates the reliance on pseudo-label generation. By virtue of the lightweight design, specifically its departure from the ubiquitous student-detector paradigm in pioneering works, the proposed DPNet delivers robust performance on base classes while achieving competitive results on target classes. This effectively balances the accuracy tradeoff inherent in previous single-branch OVD methods. To sum up, our contributions are threefold.

1) Devising a parameter-efficient adaptation strategy to bridge the gap between pretrained models and downstream tasks. This is achieved through a lightweight MSA that activates hierarchical features within the frozen backbone while minimizing trainable parameters. Building upon this, a COPG network is employed to generate high-quality proposals, which is critical for accurately localizing potential target objects.
2) Formulating a novel decoupled detection paradigm to resolve the inherent conflict between knowledge retention and target concept learning in open-vocabulary detection. By designing dedicated optimization pathways for base and target classes, our framework avoids the competitive optimization and performance degradation stemming from a single, shared architecture. This preserves the high recognition fidelity on base classes while effectively learning to detect new targets.
3) Proposing DPNet, the first end-to-end dual-path network for open-vocabulary oriented object detection in remote sensing imagery. Extensive experiments on two mainstream aerial benchmarks validate that DPNet not only sets a new state-of-the-art (SOTA) in zero-shot detection but also achieves a robust detection of target instances without sacrificing performance on base categories.

II. Related Work

A. Oriented Object Detection in Aerial Images

Benefiting from the release of remote sensing datasets annotated with oriented bounding boxes (OBBs), the development of oriented object detectors has attracted growing attention [30], [31], [35], [36], [37], [38], [39]. Most of the early oriented object detectors are directly migrated from natural images or text detection by simply adding an angle prediction branch [40], [41]. However, these detectors, which have not sufficiently considered the characteristics of remote sensing images, do not obtain satisfactory detection accuracy. Therefore, a large number of tailored oriented object detectors for remote sensing images have been developed [42], [43], [44]. The RoI transformer [30] constructs RoI learner to transform horizontal RoI into oriented RoI to alleviate the problem of massive redundant computation caused by RRPN [41]. Gliding vertex [45], oriented R-CNN [31], and QPDet [46] creatively propose novel representations of oriented bounding box to further optimize the regression routes. However, all of the above methods use a two-stage detection framework, which lacks optimization in terms of detection speed. AOPG [47] and S2ANet [48] use a refined-stage detection pipeline to achieve a better balance between accuracy and speed. Recent studies have further explored remote sensing object detection under limited supervision and domain shift, such as ALS teacher [49] for semi-supervised oriented detection and dual-teacher [50] prototype consistency for unsupervised domain adaptive detection. Different from these works within a predefined category space, our work focuses on open-vocabulary oriented

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detection, where unseen categories are recognized through vision-language alignment.

B. Open-Vocabulary Detection

Knowledge-distillation based OVD methods typically transfer the open-vocabulary knowledge from VLMs, such as CLIP [19], into conventional closed-set object detectors. ViLD [22] introduces a distillation loss between CLIP image embeddings and the detector's backbone features to transfer CLIP's visual knowledge, while utilizing the CLIP text encoder to achieve efficient classification within the detection framework. HierKD [51] introduces a global-level knowledge distillation approach that integrates instance-level knowledge distillation, enabling concurrent learning of base and target categories. OADP [52] introduces an object-aware distillation pyramid to address the issues of information loss and ineffective knowledge transfer in distillation-based OVD methods. VLPLM [53] and SAS-Det [23] implement a self-training strategy in OVD, utilizing CLIP to create pseudo labels for novel categories derived from region proposals predicted by a class-agnostic proposal generator. LLaMA-UniDetector [54] explores class-agnostic proposal generation for open-vocabulary detection, while our COPG adapts this idea to remote-sensing oriented scenarios by producing orientation-aware proposals with localization-quality scoring. Li et al. [24] initiated the investigation of open vocabulary object detection in remote sensing images and introduced CastDet, which employs a multi teacher–student model self-learning mechanism to concurrently generate pseudo labels and transfer knowledge from RemoteCLIP [55]. The above methods aim to enhance closed-set detectors for the detection of open-vocabulary words. In contrast, we employ the frozen RemoteCLIP image encoder as the foundation of our detector, directly integrating RemoteCLIP's open-vocabulary knowledge into the OVD task.

Region–text pretraining-based OVD methods focus on fine-tuning visual encoders using region–text pairs to improve the alignment of region and text features. RegionCLIP [56] employs a two-stage training strategy that consists of pretraining through the alignment of region–text pairs, subsequently transferring the obtained knowledge to an object detector. GLIPv1 [57] and GLIPv2 [58] pretrain models to align class-agnostic pseudo-boxes with words from image-associated text, utilizing human-generated detection data for fine-tuning. OVR-CNN [21] first uses BERT [59] to pretrain the faster R-CNN detector [40] on image-caption pairs and then fine-tunes the model on downstream detection datasets. In addition, PromptDet [60] and DetPro [61] meticulously design the prompt embeddings to enhance alignment with the region features. However, the exploration of open vocabulary object detection in remote sensing images remains constrained, and the field presents distinct challenges that require attention. Recently, grounding-based open-vocabulary architectures have achieved prominent success by establishing tight vision-language alignments for object-level recognition. Grounding DINO [62] inherits this grounding paradigm and integrates it with the DINO detection architecture, achieving SOTA open-set accuracy. More recently, YOLO-World [63] brings open-vocabulary capability into the real-time YOLO family via a reparameterizable vision-language path aggregation network, significantly narrowing the efficiency gap. Despite this rapid progress, extending these grounding-based detectors to oriented targets in earth observation imagery remains largely unmapped.

III. Proposed Method

This section first revisits the problem formulation of OVD. We then detail the construction of our detector framework (Section III-B), which is primarily composed of three novel components: MSA module (Section III-C), COPG network (Section III-D), and dual-path decoupling (DPD) network (Section III-E).

A. Problem Definition

The goal of OVD is to train a detector $\mathcal{D}$ that can generalize to detect objects from unseen target categories $\mathcal{C}_t$.

1) Training Phase: The detector is trained on a dataset where only a set of base categories $\mathcal{C}_b$ have instance-level annotations, $\mathcal{A}_b = \{(b_i, c_i) \mid c_i \in \mathcal{C}_b\}$. The target categories $\mathcal{C}_t$ are disjoint from the base categories, i.e., $\mathcal{C}_b \cap \mathcal{C}_t = \emptyset$.

2) Inference Phase: The detector must detect objects from a combined set of categories $\mathcal{C}_{\text{inf}} = \mathcal{C}_b \cup \mathcal{C}_t$. Formally, given an input image $\mathcal{I}$ and the set of text descriptions (the vocabulary) $\mathcal{V} = \{\mathcal{T}_c \mid c \in \mathcal{C}_{\text{inf}}\}$, the detector $\mathcal{D}$ is expected to output P

$$\mathcal{D}(\mathcal{I}, \mathcal{V}) \rightarrow P = \{(b_k, c_k, s_k)\}_{k=1}^{N}$$

where b_k is a predicted bounding box, $c_k \in \mathcal{C}_{\text{inf}}$ is its associated category label, and s_k is the confidence score. The model's performance is evaluated on its ability to accurately detect objects from both the base set $\mathcal{C}_b$ and the target set $\mathcal{C}_t$.

B. Method Overview

Fig. 2 shows the pipeline of our DPNet. We build upon the faster R-CNN-O [40] framework but fundamentally rearchitect its core to resolve the critical challenges of OVD in remote sensing. Crucially, we opt for RemoteCLIP as our foundation VLM rather than standard CLIP due to its domain-specific superiority. Unlike standard CLIP, which is heavily constrained by web-harvested natural images from horizontal perspectives, RemoteCLIP is purpose-built for earth observation via pretraining on large-scale remote sensing benchmarks. By natively maintaining fine-grained visual-semantic alignment for geographic attributes, RemoteCLIP significantly alleviates the domain gap induced by overhead perspectives and complex land-cover noise. To fully unleash its capability, our framework introduces three synergistic innovations.

1) *A Lightweight MSA:* To preserve the VLM's powerful generalization, we freeze its visual and text encoders. Our parameter-efficient adapter addresses the resulting architectural mismatch. It serves as a bridge, generating a task-specific, hierarchical feature pyramid from the coarse, frozen VLM features, making them suitable for dense detection.
2) *A COPG Network:* Operating on these hierarchical features, this network is trained to identify all potential

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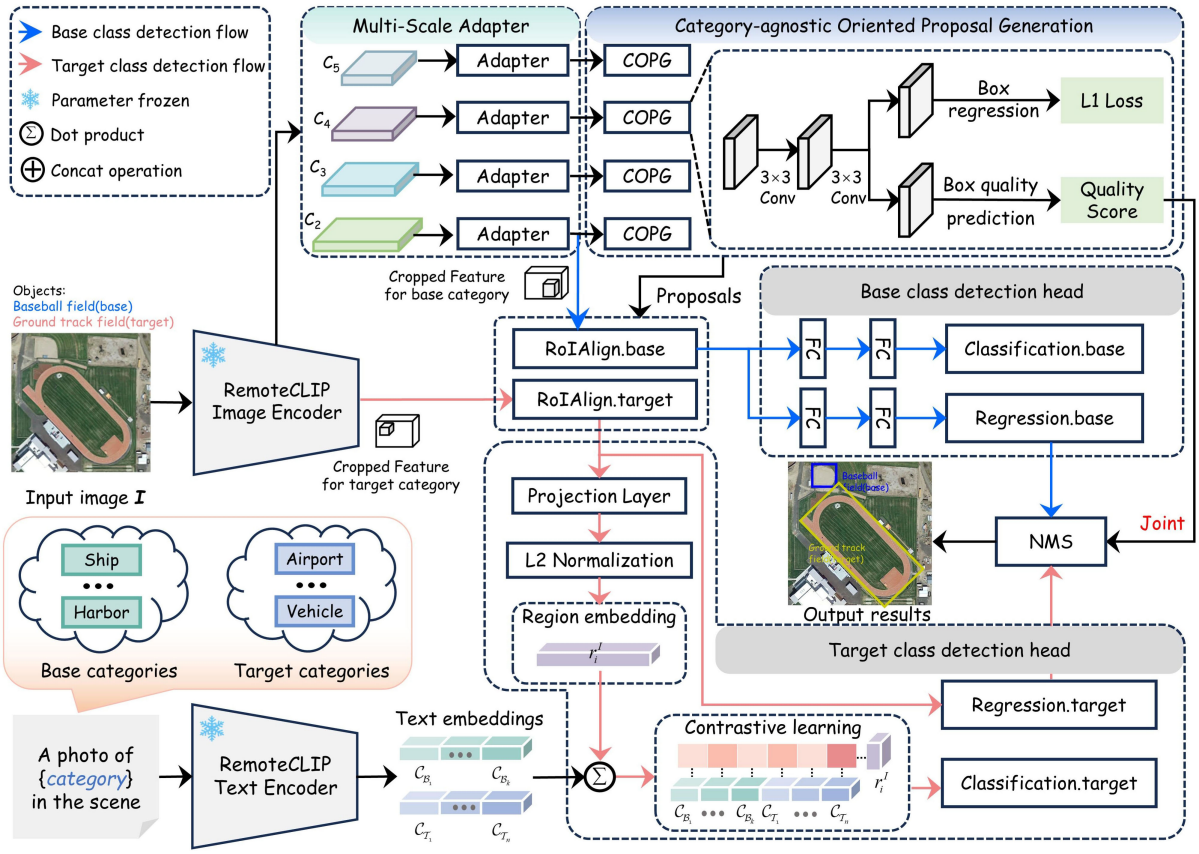


Fig. 2. Overall architecture of our proposed DPNet. A lightweight MSA first generates a hierarchical feature pyramid, on which a COPG network produces high-quality, class-agnostic proposals. Finally, our DPD network processes these proposals: the base-class path (blue flow) uses the adapter-tuned features for supervised detection, while the target-class path (red flow) bypasses the adapter, extracting features directly from the original frozen encoder to preserve general semantics. This decoupled design resolves optimization conflicts and prevents catastrophic forgetting of base categories.

objects, irrespective of their class. This design explicitly mitigates the RPN's inherent bias against unseen objects, resolving the critical bottleneck of low recall for target categories.

3) *A DPD Network:* By using dedicated, noninterfering pathways for base ($\mathcal{C}_b$) and target ($\mathcal{C}_t$) classes, the detection head resolves the optimization conflict, robustly preventing catastrophic forgetting and eliminating the performance tradeoff inherent in single-branch methods.

In Sections III-C–III-E, we will provide a comprehensive description of these innovative modules, following the network's data-flow progression.

C. Multiscale Adapter

Adapting VLMs for downstream tasks typically involves freezing the pretrained encoder [64]. This nonnegotiable strategy is critical not only for preserving the VLM's robust cross-modal alignment but also, by extension, for safeguarding its core generalization capability. Unfortunately, it exposes a severe limitation: VLM backbones, having been optimized with image-level categorical supervision, inherently struggle to produce hierarchical fine-grained representations [65]. Yet, oriented object detection in remote sensing scenarios must effectively accommodate drastic scale variations, where tiny objects (e.g., small vehicles) often coexist with massive structures (e.g., airport runways) [66]. Therefore, directly applying the VLM's coarse, single-scale features creates a profound architectural mismatch, crippling the detector's ability to accurately localize objects and resolve intricate spatial details.

To this end, we devise a lightweight MSA, serving as a parameter-efficient bridge to establish hierarchical representations without polluting the invaluable, frozen weights of the VLM. Different from standard adapters that tune a shared VLM representation for a single downstream task, MSA is designed under the dual-path constraint of open-vocabulary oriented detection. It should activate dense and scale-aware localization cues from the frozen RemoteCLIP encoder for oriented proposal generation and base-class detection, while avoiding direct disturbance to the original vision-language semantic space required by target-class recognition. Therefore, MSA is routed only to the COPG and base-class path, whereas the target-class path bypasses MSA and extracts region features from the frozen encoder. In this sense, the MSA is not a generic VLM fine-tuning adapter, but a path-specific adaptation component that enables dense localization while preserving open-vocabulary semantic alignment.

Concretely, the MSA extracts intermediate feature maps (C_2, C_3, C_4, C_5) from multiple layers of the frozen vision encoder, with each scale-specific component operating on a distinct level of spatial granularity. To ensure inference efficiency, each adapter module consists solely of a single

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1×1 convolutional layer. To prevent small objects from being mathematically obliterated, we employ a top–down modulation mechanism to facilitate cross-scale interaction. This process dynamically aggregates the high-level contextual semantics from coarser layers with the fine-grained structural details from finer layers. Specifically, the upsampled features from the coarser-level C_{i+1} are concatenated with those from the finer-level C_i to deliver P_i, the representation for the current hierarchy

$$P_i = \text{Adapter}\left(C_{i+1}, C_i\right). \tag{1}$$

In essence, this lightweight adaptation strategy successfully unlocks the latent hierarchical potential of the frozen VLM, transforming coarse embeddings into a highly adaptive feature pyramid. Consequently, it provides the critical multiscale representations required for our subsequent proposal generation and decoupled detection. The architectural choices underpinning this design are rigorously validated in the comprehensive ablation studies presented in Section IV-F.

D. Category-Agnostic Oriented Proposal Generation

As critically identified in Section I, the conventional RPN, biased by its closed-set training paradigm, systematically dismisses target objects as background. This semantic bias severely cripples the recall of novel categories, forming a fundamental bottleneck in open-vocabulary detection [24]. This issue is further exacerbated in aerial imagery, where objects with arbitrary orientations and dense packing configurations make high-quality proposal generation inherently challenging.

To directly resolve this bottleneck, we devise a COPG network. Departing from the prevalent *object-or-background* binary classification—which inherently impedes generalization—our COPG adopts a nondiscriminative, localization-centric learning objective. This design explicitly eliminates the classification bias at the proposal stage, thereby unlocking the model's capacity to encompass potential objects from all categories, seen or unseen. Concretely, the COPG operates on the hierarchical features P_i delivered by our MSA. For each level, the feature map is processed by a shared convolutional layer, followed by two dedicated branches: one for oriented bounding box regression and the other for predicting localization quality.

To accommodate the arbitrary orientations of objects in aerial imagery, we parameterize proposals using OBBs, denoted by (x, y, w, h, θ). The regression branch is tasked with predicting the refined offsets $(\delta_x, \delta_y, \delta_w, \delta_h, \delta_\theta)$ between preset anchors and their matched ground-truth boxes. Crucially, we replace the binary classification score with a localization quality score. This score is defined as the intersection-over-union (IoU) between an anchor box B_a and its corresponding ground-truth box B_{gt}

$$Q = \frac{\left|B_a \cap B_{gt}\right|}{\left|B_a \cup B_{gt}\right|}. \tag{2}$$

By training the network to regress this quality measure $Q \in [0, 1]$, we circumvent the risk of foreground overfitting and foster a generalization-friendly proposal landscape. It is imperative to clarify that during the training phase, the COPG is supervised exclusively using the bounding box coordinates of the base categories ($\mathcal{C}_b$), meaning that no target/novel class images or geometric annotations are leaked.

Beyond boosting proposal recall, the localization quality score plays a pivotal role during inference. For target categories, the frozen text encoder often yields semantically broad and thus lower classification confidence, a phenomenon we term *classification uncertainty*. To mitigate this, we intuitively fuse the COPG's quality score with the final classification confidence through a multiplicative operation. This simple yet effective strategy prioritizes well-localized proposals, yielding significant performance gains as validated in Section IV-F. In essence, the COPG module is optimized via a composite loss function

$$\mathcal{L}_{\text{copg}} = \mathcal{L}_1\left(s, s^*\right) + \mathcal{L}_1\left(\delta, \tau\right) \tag{3}$$

where s and s^* are the predicted and target quality scores, respectively, and δ represents the predicted bounding box offsets. Crucially, $\tau = (\tau_x, \tau_y, \tau_w, \tau_h, \tau_\theta)$ denotes the explicit parameterized regression targets. Following the standard conventions of oriented object detection [46], the formulas for computing these target offsets from a preset anchor box $(x_a, y_a, w_a, h_a, \theta_a)$ to its matched ground-truth oriented box $(x_{gt}, y_{gt}, w_{gt}, h_{gt}, \theta_{gt})$ are specified as follows:

$$\begin{cases} \delta_x = \left(x_{gt} - x_a\right)/w_a, & \tau_x = \left(x' - x_a\right)/w_a \\ \delta_y = \left(y_{gt} - y_a\right)/h_a, & \tau_y = \left(y' - y_a\right)/h_a \\ \delta_w = \log\left(w_{gt}/w_a\right), & \tau_w = \log\left(w'/w_a\right) \\ \delta_h = \log\left(h_{gt}/h_a\right), & \tau_h = \log\left(h'/h_a\right) \\ \delta_\theta = \theta_{gt} - \theta_a, & \tau_\theta = \theta' - \theta_a. \end{cases} \tag{4}$$

The COPG, therefore, serves as a cornerstone of our framework, ensuring high-recall, orientation-aware proposal generation that is indispensable for effective open-vocabulary detection in remote sensing imagery.

E. DPD Network

As aforementioned, the conventional parameter-sharing design introduces a critical optimization conflict between base and target classes, culminating in catastrophic forgetting of base categories [28], [29]. This conflict is particularly acute in remote sensing scenarios, where fine-grained interclass distinctions (e.g., between *bridge* and *overpass*) demand highly specialized feature representations.

Hence, we propose a DPD network, which explicitly separates the detection processes for base and target classes into dedicated, noninterfering pathways. By eliminating shared parameters between the two tasks at the detection head stage, our design ensures that the well-established knowledge for base categories remains intact while effectively acquiring new knowledge for target categories, thereby achieving a robust performance balance.

1) Base-Class Detection Path: This path is meticulously optimized for detecting base classes $\mathcal{C}_b$. It leverages the hierarchical features from the MSA and the high-recall proposals from the COPG network. Specifically, it employs *RoIAlign.base* to extract $7 \times 7 \times 256$ RoI features, which

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are subsequently processed by two parallel fully connected layers for precise bounding box regression and classification. Crucially, this pathway fully utilizes the instance-level annotations of base classes to learn highly discriminative feature distributions, thereby preserving and even enhancing the detection accuracy for known categories.

2) Novel Class Detection Path: This path is exclusively dedicated to detect unseen target classes $\mathcal{C}_t$. To prevent features fine-tuned on base classes from biasing novel concept recognition, this branch bypasses the MSA and extracts features directly from the final layer of the frozen image encoder, which retains the most generalizable semantic priors. Concretely, all proposals generated by the COPG are mapped onto this frozen layer. Since the pretrained RemoteCLIP-RN50 visual encoder utilizes a flexible ResNet backbone rather than a resolution-constrained vision transformer (ViT), it natively accommodates varying input dimensions, allowing us to resize remote sensing scenes to 800 $\times$ 800 pixels to safeguard fine-grained details. Subsequently, *RoIAlign.target* is applied to extract unified 7 $\times$ 7 $\times$ 1024 RoI features, ensuring that all dynamically generated regions are bound to a consistent spatial dimensionality required by downstream layers. These region features are then projected and normalized into visual embedding vectors.

Simultaneously, the frozen text encoder generates text embeddings for candidate categories. Specifically, a single fixed prompt template, "a photograph of {category} in the scene," is deployed to efficiently embed geographical context while bypassing the computational latency of multiprompt ensembling. To eliminate redundant textual inference overhead during testing, these text embeddings for the entire vocabulary $\mathcal{V}$ are precomputed offline once and cached as a fixed classifier matrix. Ultimately, the final prediction score is derived via a simple matrix dot-product between the runtime visual embeddings and the precached text embeddings. This divide-and-conquer design successfully leverages the robust cross-modal alignment of the pretrained VLM for zero-shot recognition while completely avoiding the catastrophic forgetting of base categories. The superiority of this feature selection strategy is empirically validated in Section IV-C.

3) Loss Function: The novel-class detection path operates in a zero-shot manner, requiring no training. Therefore, the loss function for the dual-path network is solely derived from the base-class detection path. The bounding box regression employs smooth L1 loss ($\mathcal{L}_{\text{reg}}$) and the classification employs cross-entropy loss ($\mathcal{L}_{\text{cls}}$). The overall objective is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{copg}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} \tag{5}$$

where λ_{reg} and λ_{cls} are the balancing coefficients, both set to 1.0 in our experiments. This streamlined loss design underscores the architectural elegance of our decoupled paradigm, which effectively manages the learning of both base and target classes without complex multitask objectives.

IV. Experiments and Results

A. Experimental Setup

1) Datasets: To validate the efficacy of our proposed DPNet, we conduct extensive evaluations on two mainstream aerial image benchmarks: DIOR [14] and DOTA-v1.0 [17]. Both datasets are partitioned into disjoint sets of base and target categories to construct the open-vocabulary scenario, with the specific split detailed in Table I.

TABLE I
Composition of Datasets of Base and Target Classes

Datasets	Base	Target
DIOR	APL, VE, SH, BR, OP, DAM, HA, BF, TC, GF, STA, ESA, ETS, TS, CH, STO	APO, BC, GTF, WM
DOTA-v1.0	PL, SH, HA, BR, SV, LV, BD, GTF, BC, RA, ST	TC, HC, SBF, SP

2) Implementation Details: All experiments are implemented with PyTorch 1.10.1 and MMRotate 1.0.0 on an NVIDIA RTX 3090 GPU. The model is optimized using SGD with a momentum of 0.9, a batch size of 4, a learning rate of 1 $\times$ 10^{-4}, and a weight decay of 1 $\times$ 10^{-4}. The default training schedule is 12 epochs. In our implementation, we adopt the RemoteCLIP-RN50 model,[1] with both encoders frozen during training. For textual encoding, category embeddings are precomputed offline using the fixed prompt template "a photograph of {category} in the scene" and cached on the GPU. For visual encoding, input images are resized to 800 $\times$ 800 and passed through the ResNet-50 stem and four residual stages (layer1–layer4) to produce multiscale features. We retain the original attention pooling (`attnpool`) module to project spatial feature maps into the aligned vision–language embedding space.

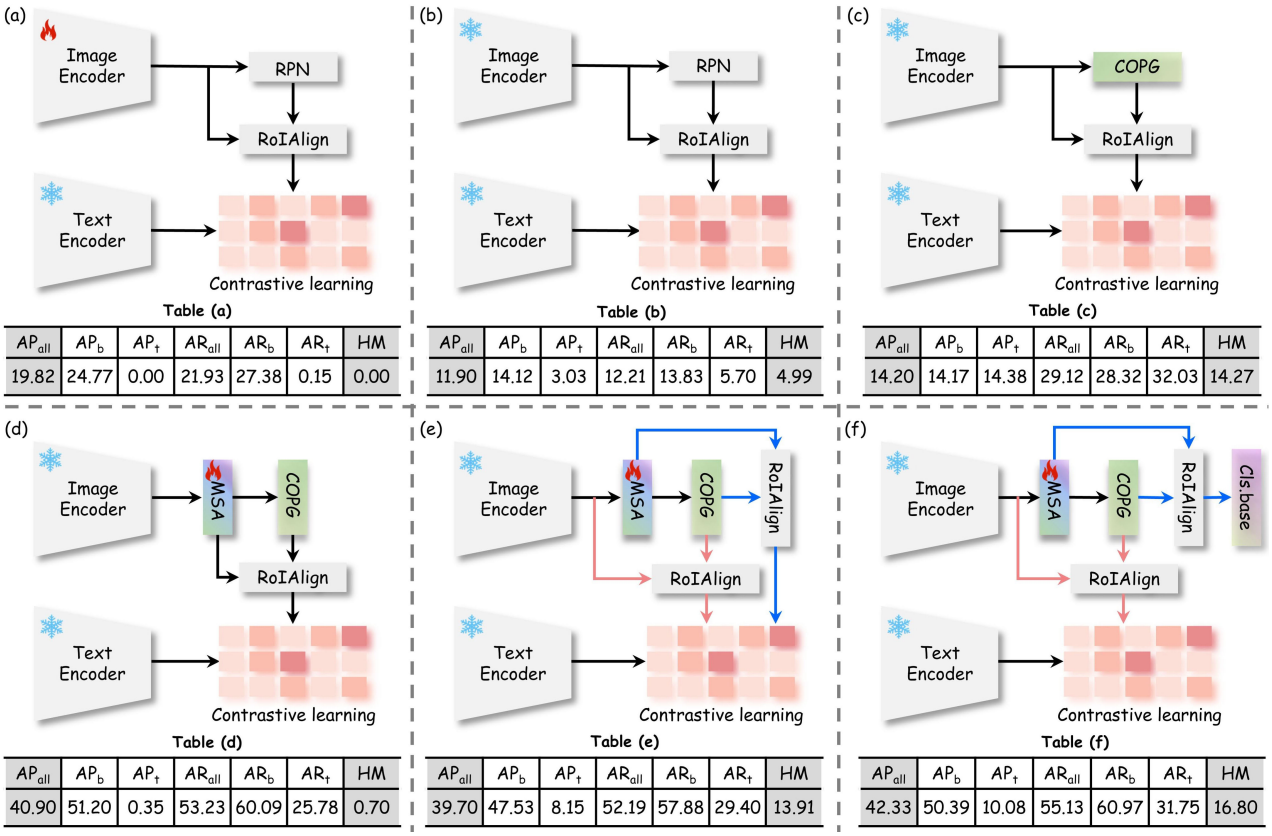
3) Reproduction of Compared Methods: For YOLO-World and OVA-DETR, we adopt their official implementations and integrate them into our MMRotate-based pipeline, built upon MMDetection, to support oriented bounding-box prediction. The adaptation is restricted to the orientation-related prediction and postprocessing components, while their original visual and language encoders and core cross-modal alignment mechanisms remain unchanged. All reproduced methods use the same dataset split, input resolution, base–target category partition, textual vocabulary, prompt template, and evaluation protocol as DPNet. Each method is trained independently on DIOR or DOTA using only the corresponding base-category annotations, without target-category supervision.

YOLO-World is initialized from its official pretrained weights and fine-tuned on the base categories for 12 epochs. Its filtering, confidence threshold, and NMS settings are kept identical to those in the official MMYOLO implementation. OVA-DETR is initialized from its official pretrained configuration, which originally adopts mixed training on DIOR, DOTA, and xView. We modify it to independent single-dataset training and train it for 12 epochs, while retaining its original NMS-free inference strategy that directly outputs the top 500 queries.

4) Evaluation Protocol: The standard mean average precision (mAP) is adopted for performance evaluation. To

[1]The pretrained RemoteCLIP-RN50 weights are available at: https://huggingface.co/chendelong/RemoteCLIP/tree/main

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AP_{all}	AP_b	AP_t	AR_{all}	AR_b	AR_t	HM
19.82	24.77	0.00	21.93	27.38	0.15	0.00

AP_{all}	AP_b	AP_t	AR_{all}	AR_b	AR_t	HM
11.90	14.12	3.03	12.21	13.83	5.70	4.99

AP_{all}	AP_b	AP_t	AR_{all}	AR_b	AR_t	HM
14.20	14.17	14.38	29.12	28.32	32.03	14.27

AP_{all}	AP_b	AP_t	AR_{all}	AR_b	AR_t	HM
40.90	51.20	0.35	53.23	60.09	25.78	0.70

AP_{all}	AP_b	AP_t	AR_{all}	AR_b	AR_t	HM
39.70	47.53	8.15	52.19	57.88	29.40	13.91

AP_{all}	AP_b	AP_t	AR_{all}	AR_b	AR_t	HM
42.33	50.39	10.08	55.13	60.97	31.75	16.80

Fig. 3. Progressive validation of our architecture components. We illustrate the step-by-step build-up from (a) trainable backbone baseline and (b) frozen backbone baseline, to (c) adding the COPG, (d) subsequently adding the MSA, (e) introducing feature-level decoupling, and finally (f) our full DPNet with decoupled heads. In (e) and (f), red/blue lines denote the target/base-class paths, respectively; black lines indicate the shared processes. All bounding box regression branches are omitted for clarity.

comprehensively assess the model's balance between base- and target-class performance, we additionally employ the harmonic mean (HM), defined as

$$\mathrm{HM} = 2 \cdot \frac{\mathrm{mAP}_{\mathrm{base}} \cdot \mathrm{mAP}_{\mathrm{target}}}{\mathrm{mAP}_{\mathrm{base}} + \mathrm{mAP}_{\mathrm{target}}}. \tag{6}$$

The HM is adopted to evaluate the balance between base-class retention and target-class generalization. Unlike the arithmetic mean, the HM penalizes imbalanced performance more strongly, thus preventing the overall score from being dominated by only base or target classes.

B. Analysis of Region–Text Alignment Strategies

To determine the optimal paradigm for vision-language knowledge transfer, we comparatively evaluated two distinct region–text alignment strategies. The image-level alignment involves cropping region proposals into individual images for separate encoding, whereas the feature-level alignment extracts corresponding region features from the holistic feature map generated by the image encoder. As demonstrated in Table II, the latter achieves mAP scores of 37.4 and 13.3 under GT and RPN settings, significantly outperforming the image-level approach. We posit that the superiority of feature-level alignment stems from its preservation of global contextual information, which maintains consistency with the pretraining paradigm of VLMs and thereby enables more effective utilization of their learned visual-semantic relationships. Based on this conclusive evidence, we adopt the feature-level alignment strategy in our novel-class detection path, extracting RoI features directly from the frozen image encoder's feature maps to ensure effective knowledge transfer.

TABLE II
ACCURACY COMPARISON OF THE PRETRAINED REMOTECLIP FOR REGION–TEXT ALIGNMENT AT BOTH THE FEATURE AND IMAGE LEVELS

Region-Text Alignment	GT	RPN
Image-level	24.9	7.6
Feature-level	37.4	13.3

C. Progressive Validation of Architecture Components

As illustrated in Fig. 3, a step-by-step ablation study from baseline to DPNet validates the necessity of each component.

1) Baseline Analysis and Problem Identification: We start from two preliminary baselines.

1) *A Trainable Backbone With a Text Classifier:* This setup replaces the R-CNN head with a text encoder. As shown in Fig. 3(a), it yields 24.77% mAP on base classes but 0.0% mAP on target classes. This zero-AP result

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indicates that the base-supervised RPN produces inadequately localized and ranked proposals for target objects, which impairs subsequent region–text classification.

2) *A Frozen Backbone With a Text Classifier:* To leverage the VLM's generalization, we freeze the image encoder [Fig. 3(b)].

This slightly improves target accuracy but degrades base-class performance. These baselines confirm two core challenges: 1) the RPN's base-class bias causes a critical proposal quality and ranking bottleneck for target classes and 2) the frozen VLM backbone lacks task-specific adaptation, leading to poor base-class accuracy.

2) Solving the Target Recall Bottleneck: To address the recall bottleneck, we first replace the standard RPN with our COPG. As shown in Fig. 3(c), this single change dramatically improves both recall and mAP for target classes, validating the generalization capability of our class-agnostic design. However, the base-class accuracy remains significantly lower than in setup Fig. 3(a). We attribute this to the frozen backbone and the minimal trainable parameters (only the COPG head), which are insufficient for adapting to the base-class task.

3) Solving the Base-Class Accuracy Collapse: We now investigate the components designed to improve base-class accuracy while preserving target performance.

a) Impact of the MSA: We first introduce the MSA to the setup from Fig. 3(c). As expected, the results in Fig. 3(d) show a significant improvement in base-class accuracy. However, the target-class accuracy drops catastrophically (16.03% → 0.35%). This drop stems from feature-space drift rather than the ineffectiveness of MSA. Since MSA is supervised only by base-category annotations, it strengthens base-discriminative representations but shifts the adapted features away from the original RemoteCLIP embedding space. As target recognition relies on the alignment between region features and frozen text embeddings, such drift severely weakens the dot-product classifier. This explains why feature-level decoupling restores target performance by routing target RoIs back to the frozen encoder features.

b) Impact of DPD: To resolve this conflict, we introduce our dual-path design. We analyze two levels of decoupling.

1) *Feature Decoupling:* We first decouple the feature source [Fig. 3(e)]. The base path uses multiscale features, while the target path uses the original frozen encoder features. This is highly effective: base accuracy is retained, and target accuracy is substantially recovered. The remaining gap is attributable to the fact that target proposals are still generated from MSA-adapted features through COPG.
2) *Head Decoupling (Full DPNet):* We then also decouple the detection head [Fig. 3(f)]. The base path uses an R-CNN head, and the target path uses the text encoder. This final step achieves the best balance.

This step-by-step validation confirms that each of our components is essential. We also note that replacing multiscale feature extraction with a stronger adapter like PANet further boosts mAP by 1% (42.3% versus 43.2%), but we retain the simpler architecture to balance accuracy and speed. This design successfully replaces complex knowledge distillation or pseudo-labeling pipelines with a simple, end-to-end network.

TABLE III
COMPARISON WITH THE SOTA METHODS ON DIOR

	Methods	AP_b	AP_t	HM
HBB	RRFS [67]	30.9	3.4	6.1
	M-RRFS [68]	32.5	4.3	7.6
	ContrastZSD [69]	51.4	3.9	7.2
	YOLO-World‡ [63]	31.4	0.0	0.0
	OVA-DETR‡ [70]	29.9	0.0	0.0
	Ours	**58.3**	**11.8**	**19.6**
OBB	YOLO-World‡ [63]	24.2	0.0	0.0
	OVA-DETR‡ [70]	22.6	0.0	0.0
	Ours	**50.4**	**10.1**	**16.8**

‡denotes our reproduction based on the official implementations. The methods are adapted to the MMDetection/MMRotate framework for oriented bounding-box prediction and retrained under an independent single-dataset protocol; see Section IV-A3 for details.

TABLE IV
COMPARISON WITH CASTDET ON DIOR UNDER THE SEMI-SUPERVISED SELF-LEARNING PROTOCOL. DPNET-SS USES THE SAME DIOR UNLABELED SUBSET AS CASTDET

Methods	Unlabeled Data	AP_b	AP_t
CastDet [24]	✓ (DIOR Unlabeled)	36.5	46.5
DPNet	× (Zero Exposure)	50.4	10.1
DPNet-SS	✓ (DIOR Unlabeled)	**57.2**	**48.9**

D. Comparison Results on DIOR

We benchmark DPNet against SOTA methods [68], [69] on the DIOR dataset (Table III). As our work is the first to specifically address oriented OVD in remote sensing, we conduct evaluations in two distinct settings—horizontal (HBB) and oriented OBB—for a fair and comprehensive comparison. In the standard HBB setting, DPNet decisively outperforms all existing methods, achieving 58.3% mAP on base classes and, more critically, 11.8% mAP on zero-shot target classes, resulting in a superior HM of 19.6. We also reproduced recent methods like YOLO-World [63], which notably scored 0.0% mAP on target classes. This aligns with the findings reported in [54] and we attribute such failure to its training paradigm's reliance on "data leakage," where target classes are implicitly seen during multidataset training, rendering it ineffective in a strict zero-shot setting. Finally, we establish the first benchmark for the oriented OVD task, where DPNet achieves 50.4% mAP (base) and 10.1% mAP (target). These strong quantitative results are supported by qualitative visualizations in Fig. 4, which confirm DPNet's ability to generate high-quality oriented boxes for unseen target classes.

1) Semi-Supervised Comparison With CastDet: CastDet [24] adopts a semi-supervised self-learning protocol, where unlabeled DIOR images containing target-category instances are used for pseudo-label generation. To provide a protocol-aligned comparison, we construct a semi-supervised self-learning variant of DPNet, denoted as DPNet-SS, which follows the same setting as CastDet and uses the same DIOR unlabeled subset for pseudo-label-based self-learning. In contrast, the default DPNet is evaluated under a strict zero-shot setting without any target-category data exposure.

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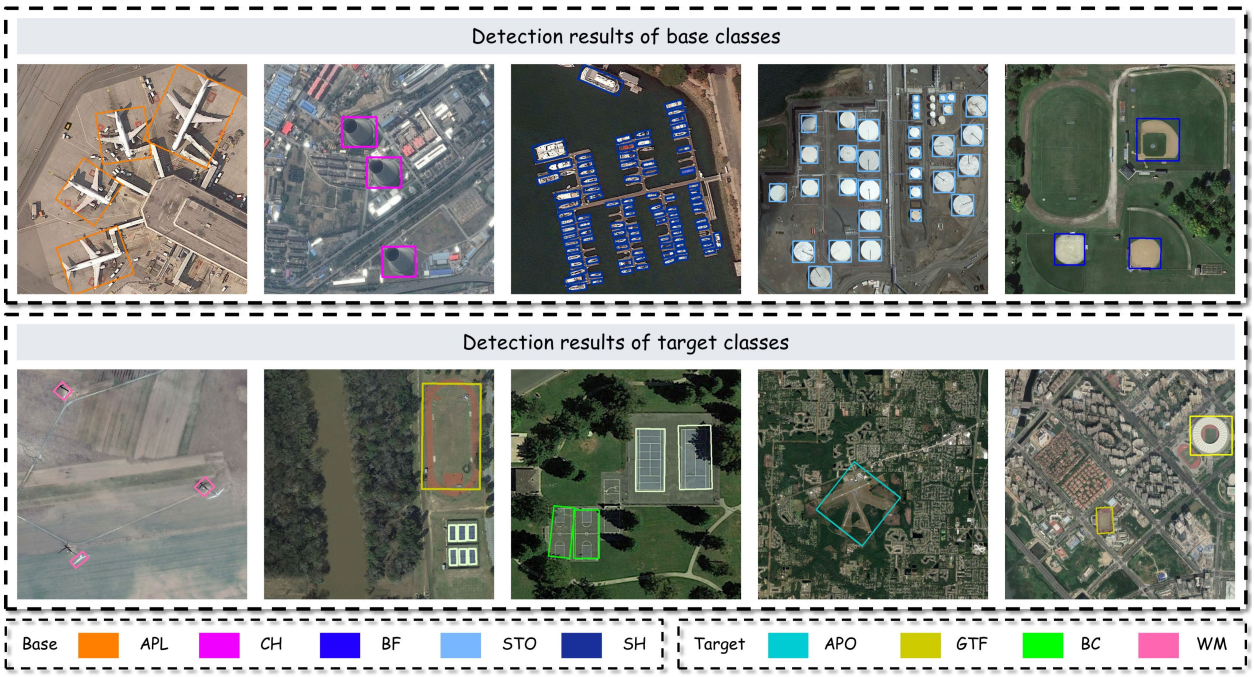


Fig. 4. Visualization results of our method for DIOR.

TABLE V
COMPARISON WITH THE SOTA METHODS ON DOTA-V1.0

	Methods	AP_b	AP_t	HM
HBB	RRFS [67]	47.1	2.2	4.2
	ContrastZSD [69]	41.6	2.8	5.2
	DescReg [71]	68.7	4.7	8.8
	YOLO-World‡ [63]	49.3	0.0	0.0
	OVA-DETR‡ [70]	46.9	0.0	0.0
	Ours	**69.6**	**20.4**	**31.6**
OBB	YOLO-World‡ [63]	47.2	0.0	0.0
	OVA-DETR‡ [70]	43.5	0.0	0.0
	Ours	**68.8**	**22.4**	**33.8**

TABLE VI
COMPARATIVE ANALYSIS OF RPN AND COPG FOR TARGET-CLASS RECALL ON THE DIOR DATASET, EVALUATED WITH 100, 300, AND 1000 PROPOSALS AT RECALL THRESHOLDS OF 0.25, 0.5, AND 0.75

	$AR_{0.25}$		$AR_{0.5}$		$AR_{0.75}$	
	RPN	COPG	RPN	COPG	RPN	COPG
100	61.6	**65.3**	37.7	**40.1**	6.5	**8.5**
300	78.7	**83.1**	54.3	**58.4**	10.0	**12.1**
1000	90.9	**93.4**	72.7	**76.2**	13.6	**16.2**

As shown in Table IV, DPNet under the strict zero-shot setting achieves much higher base-class performance but lower target-class performance, since no target-category images are exposed during training. Under the same semi-supervised protocol as CastDet, DPNet-SS reaches 48.9% AP_t, surpassing CastDet by 2.4% points while also improving AP_b by 20.7% points.

E. Comparison Results on DOTA-v1.0

We also validate DPNet on the more challenging DOTA-v1.0 dataset, with results summarized in Table V. Our method achieves top performance in the HBB setting with 69.6% mAP (base) and 20.4% mAP (target), yielding a robust HM of 31.6. In the OBB setting, DPNet establishes another strong benchmark, achieving 68.8% mAP (base) and 22.4% mAP (target), for a final HM of 33.8. The qualitative results in Fig. 5 further underscore DPNet's effectiveness, demonstrating its capability to accurately detect densely packed and arbitrarily oriented target objects, validating its readiness for complex, real-world open-vocabulary scenarios.

F. Ablation Studies

1) Effectiveness of COPG and Its Quality Score Fusion: We first validate the core contribution of our COPG: resolving the target recall bottleneck. As shown in Table VI, we compare the recall rates of COPG and a standard RPN under identical settings. The COPG consistently and significantly outperforms RPN across all proposal counts (100, 300, and 1000), confirming its superior ability to generalize unseen target objects. Although a standard RPN can geometrically cover part of the target objects when a large number of proposals are retained, proposals overlapping unlabeled target instances are assigned the background label during base-class-supervised training. As a result, these proposals cannot serve as valid foreground samples for subsequent region–text learning and are instead optimized toward the background category. In contrast, the COPG does not assign semantic or background labels to proposals and learns only category-agnostic localization quality, allowing proposals covering target objects to be retained for

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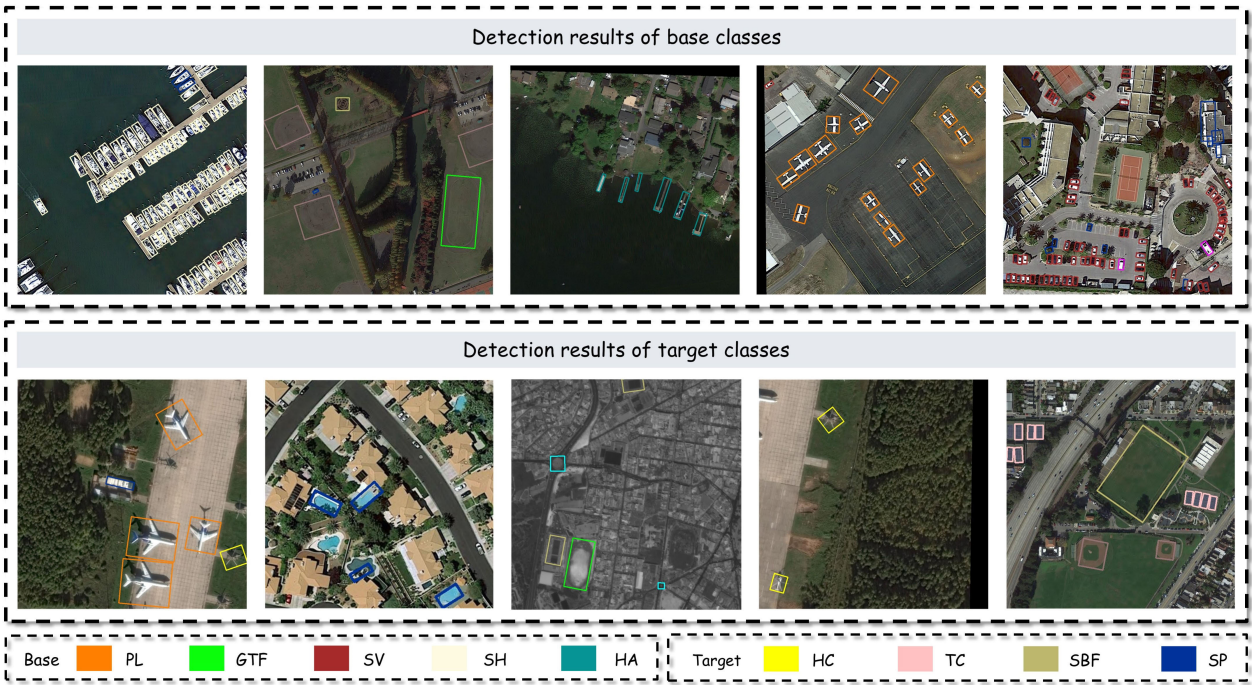


Fig. 5. Visualization results of our method for DOTA-v1.0.

TABLE VII
ABLATION STUDY OF QUALITY SCORE ON THE DIOR DATASET

Quality Score	Category	mAP
✗ / ✓	Base	9.54 / 14.17
	Target	2.70 / 14.38

subsequent detection. Moreover, as shown in Table VI, the COPG achieves consistently higher target recall than RPN under different proposal numbers and IoU thresholds, demonstrating its stronger category-agnostic proposal-generation and localization capability.

However, as observed in our initial ablation [Fig. 3(c)], high recall alone does not guarantee high mAP. Further investigation revealed a phenomenon: classification ambiguity. The VLM's general-purpose text encoder, trained on a vast corpus, assigns similar confidence scores to semantically close categories (e.g., *bridge* versus *overpass*), leading to poor discrimination. To resolve this, we multiply the COPG's localization quality score with the ambiguous classification probability. As detailed in Table VII, this simple yet effective strategy provides a substantial mAP boost, validating its critical role in disambiguating semantically similar targets.

2) Effectiveness of the MSA: The MSA module is designed to solve the base-class performance collapse caused by using a frozen, nonadapted backbone. Its effectiveness is unequivocally demonstrated by comparing Fig. 3(c) and (d): introducing the MSA increases the base-class mAP from 14.17% to 51.20%. This confirms the adapter's dual role: 1) it introduces vital, task-specific trainable parameters for the base classes and 2) it provides the multiscale feature hierarchy required for remote sensing objects.

TABLE VIII
ABLATION STUDY OF DIFFERENT MULTISCALE FEATURE EXTRACTION NETWORKS ON THE DIOR DATASET

Modules	AP_b	AP_t	mAP	Param.
FPN [73]	50.95	8.05	42.37	3.19M
PANet [74]	52.06	8.33	43.32	6.57M
MSA	50.39	10.08	42.33	0.94M

We further compare our lightweight MSA to standard FPN and PANet in Table VIII. Our MSA achieves comparable accuracy to FPN (42.33% versus 42.37% mAP) while being more parameter-efficient. More importantly, the MSA obtains higher target-class AP than both FPN and PANet, indicating that stronger feature adaptation does not necessarily benefit open-vocabulary recognition. A heavier multiscale module may improve base-class fitting, but it can also increase the risk of drifting away from the frozen VLM semantic space. Therefore, the lightweight design of MSA is a deliberate tradeoff between dense localization adaptation and semantic preservation, rather than a simplified implementation of conventional feature-pyramid modules. The superior performance of the more complex PANet (43.2%) confirms this as a promising direction for future improvement. However, to maintain an optimal balance between accuracy and efficiency, we adopt our lightweight MSA in the final DPNet architecture.

G. Portability Evaluation

We adopt faster R-CNN as our primary baseline for its simplicity, but the core components of DPNet (MSA, COPG, and DPD) are designed to be detector agnostic. To validate this

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TABLE IX
PERFORMANCE OF THE PROPOSED FRAMEWORK TRANSFERRED TO DIFFERENT DETECTORS ON THE DIOR DATASET. "SOD" DENOTES SUPERVISED OBJECT DETECTION, WHILE "OVD" REPRESENTS OVD

Methods	Task	AP_b	AP_t	mAP	HM
DODet [75]	SOD	-	-	65.10	-
	OVD	51.49	12.18	43.63	19.70
QPDet [46]	SOD	-	-	64.20	-
	OVD	51.18	11.68	43.28	19.02
Oriented R-CNN [31]	SOD	-	-	64.33	-
	OVD	51.76	10.93	43.60	18.05

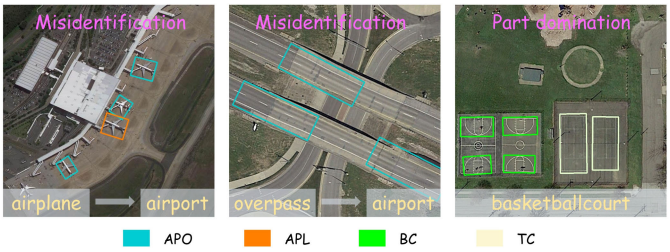


Fig. 6. Examples of failure cases evaluated on the DIOR dataset.

portability, we integrated DPNet into several classic oriented object detection frameworks, including DODet, QPDet, and Oriented R-CNN. As reported in Table IX, our method consistently delivers strong open-vocabulary performance across all architectures. All variants achieve robust and comparable results. This not only demonstrates DPNet's broad applicability and seamless integration but also implies that its performance can be further enhanced when combined with more advanced, fully supervised detector backbones.

H. Analysis of Failure Cases

We analyze typical failure cases in Fig. 6 to understand the model's limitations. Some errors reveal a high level of VLM-driven reasoning, such as in the first column where an *Airplane* (base) is misclassified as an *Airport* (target). This error, while technically incorrect, shows the model established a plausible semantic association, linking airplanes to airports. In contrast, the second column shows the overreliance on low-level visual similarity, where an *overpass* (base) is misclassified as an *airport* (target), likely due to their shared large, gray, concrete textures rather than their distinct structural features. Furthermore, the third column presents an insightful case where a *basketball court* (target) is detected as two separate half-courts. This incompleteness error implies the model has successfully learned the key discriminative features (e.g., hoops and keys) that distinguish it from visually similar categories, such as *Tennis Court*. These analyses highlight that a core, remaining challenge in OVD is mitigating confusion between semantically related and visually similar categories.

V. CONCLUSION

This article pioneers the field of open-vocabulary oriented object detection in remote sensing, proposing DPNet, a novel and efficient dual-path framework. Our work is designed to solve two fundamental challenges that arise when adapting standard OVD methods to aerial imagery: 1) the critical recall bottleneck of unknown target objects, caused by proposal networks biased toward base classes and 2) the catastrophic forgetting of base-class knowledge, caused by optimization conflicts when learning new targets. DPNet alleviates these challenges via two synergistic components. First, a parameter-efficient MSA activates hierarchical, task-specific features from the frozen VLM encoder. This empowers our COPG strategy, paving the way to eliminate the intrinsic bias of proposal network and procure high-quality proposals for all objects (seen or unseen) at varying scales. Second, our core DPD network explicitly separates the detection pathways for base and target classes. This design fundamentally mitigates knowledge interference and eliminates the destructive performance tradeoff inherent in single-branch architectures. By integrating these components, DPNet effectively balances base- and target-class detection, establishing a robust, generalizable, and efficient framework for OVD in remote sensing imagery.

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